The Counterfactual Referent Problem: A Formal Proof That Extreme-Event Attribution Claims Are Not Physical Statements, and the Documented Pattern of Their Fraudulent Presentation

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Abstract

Extreme-event attribution studies report ratios of the form $R = P(E \mid F)/P(E \mid C)$, where F denotes a factual climate-model run with anthropogenic forcing and C denotes a counterfactual run with that forcing removed. The output R is presented to the public, to courts, and to policymakers as a physical fact about the world. This paper proves that R is not a physical fact and cannot be made into one by any refinement of the models that produce it. The proof does not depend on model resolution, tuning, parameterization, ensemble spread, selection bias, or tail-distribution assumptions; it is prior to all of them. It rests on a category error. $P(E \mid C)$ has no physical referent, because the sample space of counterfactual Earth trajectories fails every operational-access condition required of a physical probability. This first result follows from Definition A.1 and from empirical premises about the state of the physical world (no laboratory analog, no analog planetary system) that we make explicit. We catalog specific published misrepresentations with authors and quotations, formally define scientific fraud in terms of misrepresentation of epistemic type, and prove that the documented pattern satisfies the definition conditional on the theorem, on the interpretive convention that unqualified counterfactual claims about physical events, quantities, or event classes refer to those physical objects, on an operational standard of professional-competence access, and on the empirical premise that the class of statements is issued in reliance contexts with material consequences. Under those four premises, the pattern is scientific fraud in the sense of Definition A.8. In the documented cases, the presenters claim to measure the effect of anthropogenic forcing on the probability of physical events. Under the theorem, they cannot.

Keywords: event attribution, counterfactual inference, climate models, philosophy of science, category error, operational access, scientific fraud, epistemology of models, World Weather Attribution (WWA)

1 Introduction

The extreme-event attribution literature, exemplified by the output of the World Weather Attribution collaboration Otto (2016) [26], the annual *Explaining Extreme Events* supplements to the *Bulletin of the American Meteorological Society* Herring et al. (2022) [15], and Chapter 11 of the IPCC Sixth Assessment Working Group I report IPCC (2021) [16], produces quantitative statements of the form:

> *Anthropogenic climate change made event E approximately R times more likely, or increased its intensity by a factor of R.*

The number R is derived from paired climate-model runs. One run, the factual run F, includes estimated anthropogenic forcing. The other run, the counterfactual run C, removes that forcing. The frequency of the event class containing E is computed in each run. The ratio of these frequencies is reported as R.

The output is presented to the public as an empirical finding McSweeney and Tandon (2024) [5]. It is cited in climate litigation Setzer and Higham (2024) [37]. It is used by insurance regulators, national climate assessments, and policy planning documents. The financial and legal consequences of these citations are substantial.

Given the model, the frequencies are what they are. This paper proves a stronger and prior claim. R is not, and cannot be, a statement about physical reality. It is a statement about the model. The category error committed in presenting R as a physical fact is independent of every technical defect of the underlying models, and would remain even if those technical defects were fully resolved.

Section 2 gives a plain-language summary of what the paper proves and how the proof works, with the formal definitions, theorem, lemma, proof, and corollaries deferred to Appendix A. Section 3 catalogs specific published misrepresentations with named authors and verbatim quotations. Section 4 proves, conditional on the theorem, on the interpretive convention that unqualified counterfactual claims about physical events refer to those physical events, on an operational standard of professional-competence access, and on the empirical premise that the statements are issued in reliance contexts with material consequences, that the documented pattern satisfies the definition of scientific fraud. Section 5 addresses the anticipated objections. Section 6 establishes that the factual-match premise on which those objections rely is itself empirically false, documented across the tropospheric-temperature comparison, individual-model performance, the statistical status of the multi-model ensemble mean, the interpretive meaning of "*match*" in the field's practice, and the extension of systematic model-observation divergence to other climate subsystems. Section 7 states the conclusion. Appendix A gives the formal proof of the theorem. Appendix B gives the formal proposition, proof, and conditionality remark for the fraud claim of Section 4.

2 Summary of the Proof

Extreme-event attribution studies report ratios of the form $R = P(E \mid F)/P(E \mid C)$. Here F names a climate-model run of Earth's history with human-caused greenhouse-gas emissions included, and C names a run of the same model with those emissions removed. The output R is presented to the public, to courts, and to policymakers as a physical fact about the world, of the form *event E was R times more likely because of human influence on the climate.*

The paper proves that R is not a physical fact and cannot be made into one. The proof does not turn on whether the models are good or bad, whether they are well-tuned or poorly tuned, whether the ensembles agree, whether the events were picked without bias, or whether the tails of the distributions are correctly estimated. All of that is beside the point. The proof rests on a category error that is prior to every one of those questions.

The category error is this. A probability, understood as a fact about the physical world, is a statement about a class of outcomes that can actually be observed. Coin flips can be observed. Mortality rates can be observed across a large population. Weather statistics can be observed over a long factual record. In every case, the number is a probability because there is a real class of realized outcomes to count.

The denominator $P(E \mid C)$ has no such class. It refers to the frequency of event E on an Earth whose industrial revolution never happened. That Earth does not exist, was never observed, and cannot be observed. It is not a laboratory system, so no experiment reproduces

it. There is no second planet with Earth's coupled atmosphere-ocean-cryosphere-biosphere dynamics but a different forcing history, so no analog population supplies the class either. No procedure applied to observations of the physical world can ever fill in the denominator, because there are no observations of the counterfactual world to apply a procedure to. Every route by which a physical probability could exist is closed. The formal statement of this fact is the Operational-Access Lemma in Appendix A.

The numerator $P(E \mid F)$ is a probability computed inside the model as well, since it is obtained by running the model many times with human-caused forcing included and counting how often event E shows up. It is a well-defined quantity inside the model. It is not a measured frequency of event E on the real Earth.

The ratio R is therefore a ratio of two model-internal quantities. It is a fact about the model, not a fact about the world. Its presentation as a physical fact identifies a model-internal quantity with a physical quantity that does not exist.

A common defense is that if the model reproduces the observed factual record well, then its counterfactual output should be trusted. The paper shows that this defense fails for two independent reasons. First, on identification: even granting the premise, reproducing the factual record does not pick out a unique counterfactual value. Two peer-reviewed attribution studies of the same 2021 Pacific Northwest heatwave, both applying accepted attribution methods, arrive at central-estimate ratios that differ by a factor of approximately 19. If reproduction of the factual record forced a unique counterfactual, this could not happen. Second, on the premise itself: Section 6 documents that the factual-reproduction premise is empirically false across the tropospheric-temperature comparison, individual-model performance, the ensemble-mean object, and multiple other climate subsystems.

The theorem is a proven mathematical statement. It is not subject to empirical refutation, and no better models, larger ensembles, faster computers, or new methods can alter it. The counterfactual Earth cannot be observed, and no computational refinement changes what can and cannot be observed. What the models compute is a fact about the model. The paper does not object to model-based reasoning stated as such; the corollaries in Appendix A give the legitimate model-scoped reading of each claim. What the paper rules out is the identification of the model output with a physical measurement of the world.

The definition of scientific fraud used in this paper turns on the misrepresentation of one kind of claim as another, together with the presenter's competence to know the difference, in a setting where others rely on the claim. Intent to deceive is not part of the definition. The full formal definitions of physical probability, model-internal probability, epistemic type, and scientific fraud, along with the theorem, the operational-access lemma, the proof, and the corollaries, are given in Appendix A.

3 Documented Misrepresentations in the Published Literature

This section catalogs specific instances of the category error in the peer-reviewed attribution literature and its official communications. In each case, the misrepresentation is quoted verbatim, attributed to named authors, and mapped to the specific step of the proof it violates. The list is illustrative, not exhaustive. Every rapid-attribution study issued by World Weather Attribution and every attribution assessment in IPCC Sixth Assessment Working Group I Chapter 11 exhibits at least one of the failures cataloged below.

3.1 Misrepresentation 1. The "virtually impossible without human-caused climate change" construction

Philip et al. (2022) [28], in the peer-reviewed rapid attribution of the 2021 Pacific Northwest heatwave, state that the observed heatwave was "*virtually impossible without human-caused climate change.*" This statement asserts that in the counterfactual world C, the event would essentially never occur. The assertion is a claim about the frequency of the event in the sample space Ω_C.

By Lemma A.13, Ω_C is operationally inaccessible, and by Corollary A.15, no physical probability is defined on it. The statement therefore asserts a physical property of an object for which no such property exists. The correct statement is: *in the model M used by the authors, integrations under the counterfactual forcing ϕ_C do not produce an event of this magnitude within the sample tested.* The peer-reviewed article contains no such qualifier attached to the "virtually impossible" claim. The claim is presented under the interpretive convention of Corollary A.18 as a physical property of Earth's climate. It is not one.

3.2 Misrepresentation 2. Directly contradictory ratios for the same event

For the same 2021 Pacific Northwest heatwave, Philip et al. (2022) [28] report that the event was "*at least 150 times less common without human-induced climate change.*" Independently, Leach et al. (2024) [18], using operational weather-prediction models, report that "*human influence on the climate made this event at least 8 [2–50] times more likely.*"

Both papers are peer-reviewed. Both concern the same physical event. Their central estimates differ by a factor of approximately 19. Their intervals do not overlap: the upper bound of the Leach et al. (2024) [18] interval (50) lies below the lower bound of the Philip et al. (2022) [28] estimate (150). Neither paper's uncertainty interval covers the other paper's central estimate.

This differential is the empirical demonstration invoked in Step 5 of the proof of Theorem A.11. If either $R(E)$ were a physical quantity, the two estimates would be measurements of the same physical quantity and would agree within their stated uncertainties. They do not. Different modeling choices, applied to the same physical event, produce different counterfactual outputs. Neither is a measurement of a physical fact. Both are model-internal quantities presented as physical facts. At least one is false as stated. On the proof, both are.

3.3 Misrepresentation 3. Directly contradictory ratios from parallel institutional attribution efforts

Zachariah et al. (2022a) [42], in a World Weather Attribution rapid analysis of the March–April 2022 India–Pakistan heatwave, report that the event was "*about 30 times more likely because of human-caused climate change.*" In the same month, the UK Met Office, using a different climate model, reported the same class of event to be "*100 times more likely*" Met Office (2022) [23].

The two estimates differ by a factor of approximately 3.3. Both were issued as institutional statements to the press. Both were presented as physical facts about the effect of anthropogenic forcing on the probability of the event. The disagreement was noted at the World Weather Attribution press briefing, at which Zachariah et al. 's coauthor Otto

stated that "*the real result is probably somewhere between ours and the Met Office result*" [6]. Neither retraction nor qualifier followed. Both numbers remain in circulation as attribution facts.

3.4 Misrepresentation 4. Numerical precision without stated model dependence

Zachariah et al. (2022b) [43], in the World Weather Attribution report on the 2022 Nigeria floods, state that the rainfall event was "*about 80 times more likely*" as a result of climate change. This statement was reproduced without qualifier by the Associated Press and other major outlets [2]. The report itself contains extensive discussion of the model ensemble and the uncertainties in the individual model outputs. The two-significant-figure ratio "*80 times*" is nevertheless presented to the public as the finding, unqualified by any statement that the number is a function of the chosen ensemble, chosen tuning targets, and chosen event definition.

Climate Central, in the attribution of Hurricane Milton on 7 October 2024, reported that the anomalously warm Gulf of Mexico sea-surface temperatures observed prior to the storm were "*400 to 800 times more likely*" due to climate change [8]. The three-orders-of-magnitude ratio was communicated to the public as a physical fact. No qualifier accompanied the numerical claim in the alert.

By Corollary A.18, communication of $R(E)$ without a model-scope qualifier presents a model-internal quantity as a physical quantity, under the default interpretive convention of scientific and journalistic communication.

3.5 Misrepresentation 5. The IPCC WG1 Chapter 11 assessment language

IPCC (2021) [16], in Chapter 11 of the Sixth Assessment Report, assigns confidence levels to attribution statements of the form: it is *virtually certain*, *extremely likely*, *very likely*, or *likely* that human influence has increased the intensity or frequency of a given class of extreme event. The confidence levels are IPCC calibrated-language terms defined as probability intervals [20].

A probability interval requires a well-defined sample space. The IPCC's calibrated probabilities are presented as expert credences over the truth of physical statements about the effect of anthropogenic forcing on event frequencies. Those physical statements are, by Theorem A.11 and Corollary A.15, not well-defined, because the counterfactual sample space fails operational access. A credence assigned to a physically undefined proposition is not a well-defined physical probability. The IPCC's use of calibrated-language terms in Chapter 11 conveys precision that the underlying claims do not possess.

The chapter authors, in the technical text, occasionally acknowledge the model-dependence of the underlying computations. The Summary for Policymakers and Technical Summary carry those results forward as statements about the physical world without the model-dependence qualifier. This is the pattern the theorem identifies as the category error.

3.6 Misrepresentation 6. Onward transmission through aggregator and legal channels

McSweeney and Tandon (2024) [5] maintains an "attribution tracker" listing hundreds of attribution studies and their headline ratios. The tracker presents the ratios as physical findings. It does not attach the model-dependence qualifier required by Corollary A.18 to any of them. Setzer and Higham (2024) [37] document the citation of attribution ratios in active climate litigation, including *City of New York v. BP p.l.c.* and *Held v. Montana*. Court filings quote the ratios as scientific facts. Neither the litigants nor the courts, on the record surveyed, apply the required qualifier.

The compounding of the original category error through these channels does not repair it. It amplifies it. Each downstream user relies on the upstream presentation. Each upstream author has published $R(E)$ without the qualifier that Theorem A.11 requires.

3.7 Misrepresentation 7. Michael E. Mann on Hurricane Harvey and general extreme-event causation

Michael E. Mann, professor of atmospheric science at Pennsylvania State University at the time of the statement and subsequently at the University of Pennsylvania, in an op-ed published in *The Guardian* on 28 August 2017 [19] under the headline "It's a fact: climate change made Hurricane Harvey more deadly," stated: "*There are certain climate change-related factors that we can, with great confidence, say worsened the flooding.*" Mann further stated: "*Harvey was almost certainly more intense than it would have been in the absence of human-caused warming, which means stronger winds, more wind damage and a larger storm surge.*"

Each claim is a counterfactual comparison: what Harvey's flooding or intensity "*would have been*" under a forcing history in which the anthropogenic component was absent. That forcing history did not occur. By Lemma A.13 and Corollary A.15, no physical quantity on the corresponding counterfactual sample space is defined against which Harvey's actual flooding or intensity can be compared. The comparisons are computed from models or estimated from thermodynamic identities applied to model reconstructions of the counterfactual world. Their public presentation as facts about the physical Hurricane Harvey misrepresents their epistemic type.

Mann has extended the pattern to a general causal claim. In an interview with David Sirota published by Sirota (2018) [39] on 24 August 2018, Mann stated: "*It's sort of in the domain of the link between smoking cigarettes and lung cancer. We don't question that link. You know, smoking cigarettes leads to lung cancer and other diseases, because it increases the likelihood of those afflictions by such a large amount that we basically equate it with a causal relationship... We would not be seeing this in the absence of climate change, so the signal of climate change as expressed in extreme weather is now undeniable... We are talking about floods, droughts, heat waves, wildfires and superstorms, you name it. We have seen unprecedented events in each of those categories over the last year that we would not have seen in the absence of climate change.*" The smoking-cancer causal chain rests on independently observed epidemiological cohorts satisfying condition (OA-3) of Definition A.1. The counterfactual attribution claim rests on a sample space that fails all three operational-access conditions (Lemma A.13). The asserted parity of epistemic status is unavailable.

3.8 Misrepresentation 8. James Hansen on the 2010 Moscow and 2011 Texas–Oklahoma extremes

James Hansen, former director of the NASA Goddard Institute for Space Studies, in Hansen et al. (2012) [12] published in *Proceedings of the National Academy of Sciences*, stated in the abstract: "*we can state, with a high degree of confidence, that extreme anomalies such as those in Texas and Oklahoma in 2011 and Moscow in 2010 were a consequence of global warming because their likelihood in the absence of global warming was exceedingly small.*" In the body of the same paper, Hansen and co-authors stated: "*the extreme summer climate anomalies in Texas in 2011, in Moscow in 2010, and in France in 2003 almost certainly would not have occurred in the absence of global warming with its resulting shift of the anomaly distribution.*"

Both statements assert counterfactual dependence of named physical events on the anthropogenic component of forcing. Each event named is an element of the realized factual trajectory. The claim that it "*was a consequence of*" or "*almost certainly would not have occurred in the absence of*" anthropogenic forcing is a claim about the counterfactual sample space Ω_C: that in Ω_C, the event class containing that event is absent or nearly so. By Lemma A.13, Ω_C fails operational access and no such absence-or-presence property is defined on it. Hansen's calibrated-language qualifiers ("*a high degree of confidence*", "*almost certainly*", "*exceedingly small*") assign numerical credences to propositions that lack defined sample spaces; by the argument of Misrepresentation 5, no such credence is a well-defined physical probability.

3.9 Misrepresentation 9. Zeke Hausfather on return-period shifts for extreme heat events

Zeke Hausfather, research scientist at Berkeley Earth, contributor to Carbon Brief, and contributing author on the IPCC Sixth Assessment Report, in a post on X (formerly Twitter) on 25 June 2021 [13], quote-posting a thread by Gavin Schmidt (see Misrepresentation 12), stated: "*This is a good way to think about climate change and extreme heat events. What was once a 1/100 year event might now be a 1/20 year event. If we keep warming the planet by the end of the century it might be a 1/5 year event. Long-term warming dramatically increases the odds.*"

The statement reports specific probability-shift ratios for extreme heat events: from a return period of 100 years to one of 20 years (a factor of 5), and prospectively to one of 5 years (a further factor of 4). Each return period is a physical-probability claim on the sample space of Earth's climate under a specified forcing regime. The "*was*" return period is defined on an earlier factual trajectory to which a sample-space stationarity assumption is implicit. The "*now*" return period compares a claimed present physical probability against the earlier one. The prospective "*end of the century*" return period compares a claimed future physical probability against both. Each of the three probabilities is defined on a sample space that either fails (OA-1) through singleton observation on a non-stationary process or, for the counterfactual and prospective conditions, fails all three operational-access conditions. The "*dramatically increases the odds*" claim compares model-internal outputs across those conditions and presents the comparison as an unqualified physical property of the class of extreme heat events.

3.10 Misrepresentation 10. Katharine Hayhoe on hurricanes and wildfires

Katharine Hayhoe, atmospheric scientist and endowed professor at Texas Tech University at the time of the statement, in an interview broadcast on NPR's *All Things Considered* on 30 August 2020 [14], stated: "*In a warmer world, hurricanes are not getting more frequent, but they are getting stronger. They're intensifying faster. They're getting bigger and slower. And they have a lot more rainfall associated with them than they would have 50 or 100 years ago.*" In the same interview, Hayhoe stated of Western United States wildfires: "*Since the 1980s, due to climate change, wildfires have burned over twice the area than they would have normally.*"

Each statement is a counterfactual comparison. The hurricane claim compares present hurricane intensity, rainfall, size, and translation speed against what those physical properties "*would have*" been on Earth's climate trajectory 50 or 100 years ago. The wildfire claim asserts a specific quantitative counterfactual: over twice the burned area compared to what "*would have*" occurred under a counterfactual condition "*normally*". Both counterfactual references specify sample spaces that fail operational access under Lemma A.13. The quantitative "*twice the area*" ratio, presented unqualified as a physical fact about wildfire behavior, is a model-internal quantity of the form Corollary A.18 identifies as misrepresented under the interpretive convention of that Corollary.

3.11 Misrepresentation 11. Kevin Trenberth on plaintiff-specific extreme weather events

Kevin Trenberth, Distinguished Senior Scientist at the National Center for Atmospheric Research, in his expert report filed in *Juliana v. United States*, No. 6:15-cv-01517-TC (D. Or.), Document 267-1, at Section 4 (Conclusions and Recommendations), page 23, stated: "*[I]t is my expert opinion that Plaintiffs including Jayden, Levi, Xiuhtezcatl, Victoria, Jaime, Journey, Zealand, and Nathan are already experiencing extreme weather events that have been exacerbated due to anthropogenic climate change*" [17, 40].

The statement attributes eight named plaintiffs' individual experiences of extreme weather to anthropogenic exacerbation as a matter of sworn expert opinion. It asserts an asymmetric causal contribution of the anthropogenic forcing component to specific realized events in the lives of named individuals. Any such asymmetric contribution requires a counterfactual reference against which the exacerbation is measured. By Lemma A.13, the counterfactual sample space Ω_C fails operational access, and no counterfactual comparison of this form has a physical referent. The reliance context is acute: the report was expert testimony disclosed under Federal Rule of Civil Procedure 26(a)(2)(B) and admissible under Federal Rule of Evidence 702, offered to establish causation for Article III standing in federal constitutional litigation, and was adopted by the court as material evidence supporting denial of the federal government's motion for summary judgment.

3.12 Misrepresentation 12. Gavin Schmidt on the attribution of specific and general weather events

Gavin Schmidt, director of the NASA Goddard Institute for Space Studies, has made repeated public statements presenting model-derived attribution claims as observed physical

facts, in venues ranging from his own bylined magazine article to his own agency's podcast to two national broadcast interviews to a technical thread on X. Five documented statements are cataloged here.

Under his own byline in *The Spectator*, 4 November 2021 [35], Schmidt wrote:

> "*The global warming we've seen so far is contributing to extreme weather worldwide. Heatwaves are becoming more frequent and more intense; rain is falling harder by about 7 per cent for every degree we warm; and when there are droughts, they are becoming more intense, with decreases in soil moisture 10 to 30 per cent greater.*"

The article appears under Schmidt's byline with the identifier "Gavin Schmidt is the senior climate adviser at [NASA]" (source has "Nasa", British publication style). The quantitative claims "7 per cent" for rain and "10 to 30 per cent" for soil moisture are ratios of a model-internal quantity under one forcing regime to the same quantity under another; neither ratio is a physical measurement. Both are presented in the unqualified form as physical facts about extreme weather worldwide.

In the NASA *Gravity Assist* podcast, 18 June 2021 [32], Schmidt stated:

> "*we're seeing, particularly with things like hurricane intensity, or rainfall intensity, drought intensity, with we're seeing these being juiced by the changes in temperature. . . . And again, that's something that's due to the warmer sea surface temperatures.*"

The verbs "*juiced*" and "*due to*" name asymmetric causal consequences of forcing on physical event classes. Any such asymmetric attribution requires a counterfactual reference against which the asymmetry is measured. By Lemma A.13, the counterfactual sample space Ω_C fails operational access, and no such physical reference is defined.

In an ABC News interview with chief meteorologist Ginger Zee, 29 September 2021 [33], released by ABC News under the headline "*NASA climate advisor: 'We're seeing things happen . . . that are clearly unprecedented'*," Schmidt stated:

> "*the fact of the matter is. . . that we can now detect the changes that. . . global warming is. . . actually having on specific weather events*"

and, in the same interview:

> "*we're seeing more category three, four, five hurricanes in the Atlantic that seems to be driven by the overall warming of the main development region, the tropical Atlantic; we're seeing that when there is drought, the droughts are more intense because there's more of an evaporative demand.*"

The claim of *detection of the changes that global warming is actually having on specific weather events* is precisely the class of claim ruled out by Theorem A.11: it asserts that a model-derived comparison across counterfactual forcing conditions has been observed in the physical world, which requires the counterfactual condition to be operationally accessible.

On *PBS NewsHour*, 6 September 2023 [36], Schmidt stated:

> "*But what we're seeing now is that those things that were one-in-100-year events are now one-in-50-year, one-in-30-year, once-a-decade, and soon conceivably could be an every-year or an every-couple-of-year event. And that's the difference. It's not that these things have never happened before. It's not that we have never had a heat wave. It's not that we have never had strong rainfall or droughts. But the frequency and intensity of these features is increasing.*"

The statement reports four ratio shifts (from one in 100 years, to one in 50, to one in 30, to once a decade, to once every year or every couple of years). Each ratio is a comparison of model-internal probabilities across model-internal forcing conditions, delivered to a national broadcast audience as an unqualified physical property of extreme weather events. Schmidt's acknowledgment in the immediately following sentences that heat waves, strong rainfall, and droughts have occurred before establishes that the events at issue are elements of the realized factual sample space Ω_F, which is precisely the condition that puts the return-period ratios on the counterfactual sample space Ω_C and inherits its operational-access failure under Lemma A.13.

In an eleven-post thread on X on 25 June 2021 [34], from an account (*@ClimateOfGavin*) that has since been deleted but archived at Thread Reader, Schmidt set out an explicit prescription for computing attribution numbers and applied it to the Pacific Northwest heatwave that had not yet occurred:

> "*Take a situation where climate change now means that we can experience an extreme that could effectively never* happen under the previous climate [how near we are to this point is not relevant for the argument]. *a v. low probability. ... What is likelihood of exceeding, say, 105°F under the previous climate (P) and the new situation (N)? The fractional attribution to climate change is then (N−P)/N. So if P=0 (the extreme was never going to happen before), the attribution to climate change is 100%. If the odds of an event have doubled, then climate change is 50% responsible etc.*"

Concerning the specific Pacific Northwest event forecast at the time of the thread:

> "*What it means for the Pacific NW is that there will be a very high attribution to climate change for the upcoming event and the exact numbers will depend on how hot it really gets. And the hotter it gets, the larger the attribution will be.*"

The formula (N−P)/N is a direct instance of the class of claim ruled out by Theorem A.11. Its terms P ("*likelihood of exceeding ... under the previous climate*") and N ("*likelihood ... in the new situation*") are probabilities defined on the counterfactual and factual sample spaces respectively. For P, the "*previous climate*" condition understood as a forcing history without the anthropogenic component is a claim on Ω_C. By Lemma A.13, no physical probability on Ω_C is defined, and the output "*climate change is X% responsible*" is not a physical fact about the event. Schmidt's prospective attribution of the Pacific Northwest event ("*very high attribution to climate change for the upcoming event*"), applied to an event before it occurred, was subsequently followed by two peer-reviewed retrospective attributions of that same event by Philip et al. (2022) [28] and Leach et al. (2024)

[18] whose central estimates differ by a factor of approximately 19 (Misrepresentation 2). The (N−P)/N framework does not resolve the disagreement, because both papers apply the framework and arrive at non-overlapping intervals.

The five statements together document repeated presentation of model-derived attribution ratios as observed physical facts, across national broadcast audiences, a magazine readership, the presenter's own agency's outreach channel, and a technical thread on X, without a model-scope qualifier in any of them. The institutional venue (the NASA GISS directorship) is invoked or identified in every case.

3.13 Misrepresentation 13. Clarke et al. (2026), WWA Report No. 92 on the Nepal Rasuwa disaster

Ben Clarke, Mariam Zachariah, Friederike Otto, and 21 co-authors, in Clarke et al. (2026) [7], published as World Weather Attribution scientific report No. 92 on 15 September 2026, apply the WWA Protocol of Philip et al. (2020) to the 26 August 2026 rock-ice avalanche and debris flood on the north face of Langtang-Lirung in Nepal, which killed at least 1,386 people, left 5,130 missing, and displaced tens of thousands. The report does not claim to have performed direct attribution of the specific event; it states in Section 5 that its results "*do not establish the magnitude of that contribution or identify a unique trigger*" and that direct event attribution "*requires additional evidence.*" The misrepresentations documented here concern the report's temperature-magnitude attributions and its risk-of-geohazards attribution, each of which rests on counterfactual comparison and each of which is presented in the unqualified form. Three of its lead authors (Zachariah, Otto, Philip as review author) are already named in Misrepresentations 1, 3, and 4, applying the same institutional protocol.

First, in a candid methodological self-description in Section 3, the authors write:

> "*In this section, we study the influence of anthropogenic climate change by comparing the expected intensity of temperatures over these domains at present with those in a 1.4˚C cooler climate without human-caused warming (based on Forster et al., 2026).*"

The report defines "*the influence of anthropogenic climate change*" by comparison to a counterfactual sample space specifying a climate history that did not occur ("*a 1.4˚C cooler climate without human-caused warming*"). This is precisely the epistemic type ruled out by Theorem A.11: a comparison across counterfactual forcing conditions, offered as a measurement of an influence on the physical world. By Lemma A.13 and Corollary A.15, no physical magnitude on the counterfactual condition is defined.

Second, in Section 3.2.2 the authors report specific temperature-magnitude attributions at the location of the slope failure:

> "*In the wider Himalayan region covering parts of northern Nepal and the Tibetan plateau, the warmest months of July and August have become approximately 1.5˚C (0.83 to 2.2˚C) warmer, while the mean annual temperature has risen by about 2.0˚C (1.0 to 2.9˚C) warmer, due to human influence on the climate. In the vicinity of the slope failure in August 2026, the warmest months of July and*

> *August have become approximately 1.5°C (0.25 to 2.8°C) warmer, while the mean annual temperature has risen by about 1.9°C (0.85 to 3.0°C) warmer, due to human influence on the climate.*"

This extends the pattern of misrepresentation to temperature-magnitude attributions at a specific slope-failure location. Each temperature increase attributed to "*human influence on the climate*" is a comparison of a model-internal quantity under one forcing regime to the same quantity under a counterfactual forcing regime that never occurred. Each 95% confidence interval attached to the attribution is a within-model statistic that describes model spread, not a measurement of an influence on the physical Earth.

Third, in the report's Conclusions (Section 6), the authors summarize:

> "*as a direct result of human-induced climate change, annual temperatures and July-August mean temperatures have risen significantly around this slope and the wider Himalayas, with increases at approximately 1-1.5 times the global mean temperature rise.*"

"*Direct result of human-induced climate change*" is causal-attribution language attached to a temperature change ("*annual temperatures and July-August mean temperatures have risen significantly*") at a specified location. The object of the attribution is the temperature change, not the occurrence of the avalanche. The claim nevertheless rests on a comparison of a model-internal quantity under one forcing regime to the same quantity under a counterfactual forcing regime that never occurred, and by Lemma A.13 and Corollary A.15 the counterfactual condition fails operational access. The unqualified physical reading of "*direct result*" as a claim about the physical Earth is unavailable.

Fourth, in the same Conclusions section:

> "*Regardless of the complexities of direct attribution for a specific event, it is clear that human-induced climate change is driving rapid and warming-driven transformation of high mountain environments in the Himalaya, which in turn contributing to the risk of potentially catastrophic geohazards, especially in regions such as Nepal with substantial seismic activity. Until emissions from fossil fuels are brought to net zero globally, this risk will continue to grow.*"

The opening clause "*Regardless of the complexities of direct attribution for a specific event*" is a disclaimer: the report elsewhere states explicitly that its results "*do not establish the magnitude of that contribution or identify a unique trigger*" and that direct attribution of the specific event "*requires additional evidence.*" The rest of the sentence and the sentence that follows do not override that disclaimer; they make a different attribution, namely of (a) "*rapid and warming-driven transformation of high mountain environments*" and (b) an increase in "*the risk of potentially catastrophic geohazards.*" Both of those are counterfactual claims in their own right. The transformation claim is a comparison of the physical state of the mountain environment under the realized forcing with its state under an anthropogenic-forcing-absent counterfactual. The risk claim is a probability statement about a class of physical events under one forcing regime against another. Both counterfactual references specify sample spaces that fail operational access under Lemma A.13. Neither claim has a

physical referent, and the immediate policy tie ("*Until emissions from fossil fuels are brought to net zero globally, this risk will continue to grow*") is directed at an audience for whom Corollary A.18's interpretive convention applies: a policymaker or citizen who reads the risk claim as a physical statement about future geohazards in Nepal.

Aggravating features of this misrepresentation: the report was issued three weeks after the event by the same WWA team responsible for Misrepresentations 1, 3, and 4, applying the same WWA Protocol; its distribution is aimed explicitly at humanitarian and policy audiences (Red Cross Red Crescent Climate Centre, Climate Analytics South Asia, the Stimson Center, IFRC, and government stakeholders in Nepal are among the listed institutional affiliations of the co-authors); the counterfactual comparisons carrying the temperature-magnitude and risk-shift attributions are presented in the unqualified form even though the report's own body explicitly disclaims direct attribution of the specific event.

3.14 Summary

Table 1: The thirteen documented misrepresentations. Claim-type key: R = frequency ratio $R(E)$; D = counterfactual dependence claim ("*would not have occurred*", "*consequence of*"); M = counterfactual magnitude claim ("*X°C warmer due to*"); T = detection claim ("*we can detect the effect of climate change on specific weather events*"). All thirteen fall within the scope of Corollary A.18.

#	Presenter	Venue	Year	Type	Key phrase
1	Philip et al. (WWA)	*Earth System Dynamics*	2022	R	"virtually impossible without human-caused climate change" (PNW heatwave)
2	Philip et al. vs. Leach et al.	*ESD* vs. *Nat. Commun.*	2022, 2024	R	150× vs. 8× [2–50]; intervals non-overlapping; same event
3	Zachariah et al. vs. UK Met Office	WWA report vs. press release	2022	R	30× vs. 100×; same event
4	Zachariah et al.; Climate Central	WWA report; attribution report	2022, 2024	R	80× (Nigeria); 400–800× (Milton), unqualified
5	IPCC WG1 Chapter 11	AR6 assessment report	2021	R	"virtually certain", "very likely" as calibrated language on undefined sample spaces
6	Carbon Brief; Setzer & Higham	attribution tracker; litigation survey	2022, 2024	R	unqualified ratio transmission through aggregator and legal channels
7	Michael E. Mann	*Guardian* op-ed; *Capital & Main* interview	2017, 2018	D	"worsened the flooding"; "in the domain of the link between smoking cigarettes and lung cancer"
8	James Hansen	*PNAS*	2012	D	"almost certainly would not have occurred in the absence of global warming" (Texas 2011, Moscow 2010, France 2003)
9	Zeke Hausfather	X (Twitter) post	2021	R	"1/100 year event might now be a 1/20 year event"
10	Katharine Hayhoe	NPR *All Things Considered*	2020	M	"twice the area than they would have normally" (wildfires)
11	Kevin Trenberth	expert report in *Juliana v. United States*	2018	D	"exacerbated due to anthropogenic climate change" (named plaintiffs)
12	Gavin Schmidt	*Spectator*; NASA; ABC News; *PBS NewsHour*; X thread	2021, 2023	R, M, T	"detect the changes… on specific weather events"; "juiced by the changes"; return-period sequence; "(N−P)/N … climate change is 50% responsible"
13	Clarke et al. (WWA)	WWA scientific report No. 92 (Nepal)	2026	M	"1.9°C warmer… due to human influence" at the slope-failure location

In every case cataloged above, the specific misrepresentation is the presentation of a model-internal ratio $R(E)$, or of a counterfactual dependence claim derived from one, as a physical fact about Earth's climate, without a model-scope qualifier. In several cases (Misrepresentations 1, 5, 7, and 8), the language goes further and asserts a physical property ("*virtually impossible*", "*virtually certain*", "*almost certainly would not have occurred*") of a sample space that fails operational access. In Misrepresentation 7, the causal claim is asserted at the parity of the smoking-cancer epistemic standard, which the counterfactual sample space cannot support. In Misrepresentation 11, the counterfactual causal claim is offered as sworn expert testimony under Federal Rule of Evidence 702 and adopted by a federal court as material evidence. In Misrepresentation 13, the pattern is extended to temperature-magnitude attributions at a specified location (with 95% "*confidence intervals*" presented as physical) and to a claim of increased risk of catastrophic geohazards tied to fossil-fuel policy, both offered in the unqualified form even though the report's own body explicitly disclaims direct attribution of the specific event. The presence of internal technical disclosures elsewhere in the same papers or in the same authors' more careful writing does not repair the category error in the statements cataloged, because those statements are communicated, cited, and relied upon in the unqualified form under the interpretive convention of Corollary A.18.

4 The Documented Pattern Constitutes Scientific Fraud

The pattern of statements cataloged in Section 3 satisfies the definition of scientific fraud stated in Appendix A (Definition A.8). The definition has three conditions. What follows walks through them in plain terms. The formal proposition, its proof, and the accompanying conditionality remark are stated in Appendix B.

The first condition asks whether the statement was presented as one kind of claim while in fact being a different, weaker kind. The statements cataloged in Section 3 are grammatically statements about the physical world. "*Virtually impossible without human-caused climate change*" names a physical event and reports a property of it. "*150 times less common without human-induced climate change*" names a ratio of physical event frequencies. "*30 times more likely because of human-caused climate change*" asserts a causal effect on the probability of a physical event class. These are presented, and are read by their intended audiences, as measurements of the physical world. They are not measurements of the physical world. By Theorem A.11 they are ratios of two model-internal probabilities, and the physical reading is not available to them. The first condition is met.

The second condition asks whether the presenter had access to the information that would establish the correct classification. The presenters divide into two groups. In the first are the authors of the specific attribution studies themselves, who ran the models, chose the parameters, specified the counterfactual forcing, and wrote the code that produced $R(E)$. They know how $R(E)$ was computed because they computed it. In the second are senior climate scientists who invoke or restate attribution claims in op-eds, media interviews, broadcast segments, expert testimony, and public lectures. Each has held a senior research position at a major climate-science institution and has published work using the same modeling and statistical machinery from which attribution ratios are derived. Each is professionally competent to recognize that $R(E)$ is a function of a model, its parameters, and its input data, and therefore is a model-internal output rather than a physical measurement. The second

condition is met by every presenter in the documented pattern.

The third condition asks whether the statement is issued in a setting where others rely on it and where reliance carries material consequences. It is. The reliance runs through three independent channels, each documented in the record. First, litigation: attribution ratios are cited in climate lawsuits with damages, statutory obligations, and constitutional rights at stake ([37]; Misrepresentation 11 documents a case in which a federal court adopted a counterfactual attribution claim as material evidence). Second, federal regulation: the US Environmental Protection Agency's Endangerment Finding under Clean Air Act Section 202(a) [11] rests on assessment reports whose content includes attribution language, and the ongoing federal debate over the finding's rescission and defense continues to frame attribution studies as decision-relevant. Third, formal scientific assessment: the National Academies commissioned a report on extreme-event attribution to inform policy [24]; the US National Climate Assessment and comparable international products cite attribution numbers directly in policy planning contexts; insurance regulators cite them in rate-approval decisions. The third condition is met.

The fraud finding depends on four premises jointly: (a) Theorem A.11; (b) the interpretive convention of Corollary A.18, that unqualified counterfactual claims about physical events refer to those physical events and not to a model's representation of them; (c) the operational standard of professional-competence access stated in Definition A.9; and (d) the empirical premise, established by the reliance-context evidence just recited, that the statements are issued in settings where others rely on them. Any counterparty who accepts these four premises is committed to the finding. Rejection of any one blocks the finding while leaving the others intact. Proposition B.1 in Appendix B is the formal statement of the finding under these four premises, together with its proof by verification of the three conditions.

4.1 Absence of intent to deceive is not a defense

Definition A.8 does not require intent to deceive. Where a presenter's misrepresentation of epistemic type is a product of professional inattention rather than deliberate misdirection, the fraud is not thereby cured. The reliance context and its consequences are unchanged by the presenter's mental state. The audience is misled to the same extent, and third parties are harmed to the same extent, whether the misrepresentation was intended or merely negligent. Cf. the treatment of fraud in professional-negligence law, where reckless disregard for the epistemic status of a representation is treated as constructive fraud regardless of proven intent.

4.2 Circumstantial evidence of intent

Intent to deceive is not required by Definition A.8 and is not deductively established by the observations available on the public record. The following facts are matters of public record and constitute circumstantial evidence bearing on intent, without amounting to a proof of it.

First, the technical text of the same papers whose public-facing statements omit the model-dependence qualifier frequently contains the qualifier. Authors who write the qualifier in Section 4 of a paper and omit it from the abstract and from the accompanying press release have made a drafting choice. Whether that choice reflects an intent to deceive, or a judgment

about the audience's needs, or an editorial convention adopted without consideration, cannot be settled from the drafting record alone. The pattern is systematic across the literature.

Second, the specific instances in Section 3 of directly contradictory ratios for the same event, issued by parallel institutional efforts (Misrepresentation 3), were the subject of contemporaneous discussion among the responsible authors. Zachariah et al. 's coauthor Otto stated at the World Weather Attribution press briefing that "*the real result is probably somewhere between ours and the Met Office result*" [6]. The quotation is on the record. Neither number was retracted after the remark; both remain in circulation as published attribution ratios. What inference to draw from the retention is a further question that the record alone does not settle.

Third, the pattern documented in Section 3, Misrepresentation 5 for IPCC Chapter 11, in which model-dependence is acknowledged in the underlying assessment text and dropped from the calibrated-language statement carried forward into the Summary for Policymakers, is not accidental. The Summary for Policymakers is drafted, negotiated, and edited by the same author community that produced the underlying assessment. The dropping of the qualifier is a deliberate editorial act. Whether it reflects intent to deceive the policymaker audience or a judgment that the qualifier is unnecessary at that level of abstraction cannot be settled from the editorial record alone.

The three facts together are not a proof of scienter. They are circumstantial evidence that a party bringing an intent-based claim would be entitled to develop. The fraud analysis under Definition A.8 does not rely on such development.

5 Responses to Anticipated Objections

5.1 Objection 1. *The counterfactual is well-defined as a probability over model realizations*

Granted. This is Corollary A.18. If the attribution community restricts its claims to the model-internal form, the theorem imposes no further burden. The theorem is directed at the transition from the model-internal statement to the physical statement. That transition is unjustified, and Corollary A.18 states the acceptable retreat position. The observed practice of the attribution literature and its downstream communicators does not adopt this retreat position. The public, courts, and regulators are told that $R(E)$ is a physical fact [5, 37]. It is not. Section 3 documents specific instances. Section 4 shows that, conditional on the theorem, on the interpretive convention of Corollary A.18, on Definition A.9, and on the empirical reliance-context premise, the pattern satisfies Definition A.8.

5.2 Objection 2. *The counterfactual is validated indirectly by the model's ability to reproduce the factual record*

Refuted twice over. First, by Step 5 of the proof of Theorem A.11 and by the empirical evidence in Misrepresentation 2 (Section 3): even granting the antecedent arguendo, reproduction of F constrains the parameter vector to the set Θ_F but does not select a unique counterfactual output. The disagreement between Philip et al. (2022) [28] and Leach et al. (2024) [18] on the 2021 Pacific Northwest heatwave, both of which are claimed to reproduce

the factual record within their respective tolerances and both of which produce counterfactual outputs differing by a factor of approximately 19, is a direct empirical demonstration. Second, and independently, by Section 6: the antecedent is empirically false. The models do not reproduce the physical record in any physical sense; the comparison target invoked by the literature is not a physical quantity; the ensemble-mean object invoked to smooth over model dispersion is not a statistical estimator; and the tolerance under which "*match*" is claimed is not a physical match. The objection fails at both ends.

5.3 Objection 3. *Multiple independent models converge on similar R, therefore R is robust*

Convergence, when it occurs, reflects shared parameterizations, shared tuning targets, shared data assimilation histories, and shared theoretical assumptions across the modeling community. Convergence under shared priors is not evidence of physical correspondence. The empirical claim of convergence is also not sustained. Angélil et al. (2018) [1], using four AGCMs across seven spatial scales, find that "the sensitivity of attribution statements to the model used is often greater than the sensitivity to a doubling or halving of the spatial scale of the event." Probability ratios (PRs) for hot tropical extremes span ∼12 to ∼250 across the four models (a factor of ∼21); PRs for cold tropical extremes span ∼0.008 to ∼0.15 (a factor of ∼19). For wet events the disagreement is categorical: "Statements do however vary largely between AGCMs in terms of whether they are positive (PR $>$ 1) or negative (PR $<$ 1)." Models disagree on the direction of the anthropogenic effect, not merely its size. Specific documented cases exceed a factor of nineteen (Misrepresentation 2) and a factor of three between two institutional efforts on the same event (Misrepresentation 3).

5.4 Objection 4. *The models are grounded in physics (radiative transfer, thermodynamics)*

Granted for the components in question. The attribution ratio $R(E)$ is not determined by radiative transfer or thermodynamics alone. It is determined by the joint behavior of resolved dynamics, parameterized convection, parameterized cloud microphysics, parameterized aerosol interactions, and parameterized ocean mixing. None of these parameterizations is derived from first principles. Each is a modeling choice. The category error established in the theorem is prior to this distinction. Even a model whose components were entirely first-principles physics would produce a counterfactual output on a sample space that fails operational access.

5.5 Objection 5. *Attribution provides useful guidance for policy even if its epistemic status is contested*

This is a retreat from a scientific claim to a policy claim. The theorem addresses the scientific claim. If the attribution community concedes that $R(E)$ is not a scientific statement about the world but a policy-guidance instrument, the theorem imposes no further burden. That concession is not currently being made. The public, litigants, courts, and regulators are told that $R(E)$ is a scientific fact.

5.6 Objection 6. *The argument would invalidate all counterfactual reasoning in science*

It would not. In fields where the counterfactual is bounded by strong theoretical constraints and by repeated observation of analogous systems, the counterfactual sample space satisfies condition (OA-3) of Definition A.1. Particle-physics null distributions are computed from Standard Model calculations that have been tested against millions of independent collisions; each such collision is an operationally-accessible member of the sample space in question. Randomized clinical trials place a treatment group and a control group in the same operational sample space by design. In each case, condition (OA-2) or (OA-3) is satisfied by construction. The Earth's climate system has exactly one realization and no independently-observed analog Earths. The counterfactual sample space Ω_C therefore fails operational access uniquely, and Lemma A.13 applies in a way it does not apply in experimentally repeatable sciences.

5.7 Objection 7. *The characterization of the pattern as "fraud" is inflammatory and unjustified*

Definition A.8 is stated in terms neutral to inflammatory usage. Its three conditions are matters of fact conditional on Theorem A.11, the interpretive convention of Corollary A.18, Definition A.9, and the empirical reliance-context premise, and Proposition B.1 verifies them under those four premises. The word *fraud* is the correct term for a statement whose epistemic type is materially misrepresented in a reliance context, by a presenter with professional-competence access to the corrective information. That the correct term is unwelcome to those it names does not license substituting a weaker one.

5.8 Objection 8. *This argument has been considered and addressed in the philosophy of science literature*

Adjacent arguments have appeared. Oreskes et al. (1994) [25] established that numerical models of open natural systems cannot be verified or validated in the strong sense. Winsberg (2018) [41] and Parker (2020) [27] have examined the epistemic status of climate ensemble outputs. The specific category error identified here, that the counterfactual sample space fails operational access and therefore admits no physical probability, has not to our knowledge been given a formal proof in the climate-attribution context. This paper supplies the proof under the operational-access premises made explicit in Appendix A, documents the pattern of misrepresentation the proof identifies, and establishes that the pattern satisfies Definition A.8 conditional on the theorem, on the interpretive convention of Corollary A.18, on the professional-competence standard of Definition A.9, and on the empirical reliance-context premise underlying condition (iii).

6 The Factual-Match Premise Is Empirically False

Step 5 of the proof of Theorem A.11 grants, arguendo, the defense's premise that model M reproduces the observed factual record, and shows that even under that premise no unique counterfactual output is selected. The premise is separately false. This section states

its empirical failure in four independent respects. Each is fatal on its own; together they establish that the "*model reproduces the factual record*" language on which the attribution literature routinely relies is, in the physical sense, not a true statement about any published CMIP-based attribution study.

6.1 The comparison target has no physical referent

Attribution studies do not compare model integrations to Earth's physical climate. They compare model output to derived scalar indices, most commonly global mean surface temperature anomaly (GMST) or regional analogs constructed on the same principle. GMST is computed as an area-weighted arithmetic mean of intensive-variable surface temperature readings from thermally disconnected grid cells. Temperature is an intensive property. Intensive-variable quantities do not admit additivity under set union, and a weighted arithmetic mean of temperature readings taken at points not in thermodynamic contact has no thermodynamic referent (Physical Tether Theorem, [9]). A GMST anomaly time series is a derived numerical index. It has no thermodynamic units in the physical sense, and it does not describe a physical state of Earth's climate system. Agreement between a model's GMST index and the observed GMST index is agreement between two derived numbers, neither of which is a physical measurement. It is not agreement between the model and physical reality.

6.2 Individual models do not match the derived index

Even taking the derived index at face value, individual CMIP integrations do not agree with the observed index. McKitrick and Christy (2020) [22] examine hindcast trends from 38 CMIP6 models against satellite, radiosonde, and reanalysis observations of the global and tropical lower- and mid-troposphere over 1979–2014 and report that every one of the 38 model runs overpredicts warming in every target observational analog, in most cases significantly. McKitrick and Christy (2018) [21] report the same directional result for 102 CMIP5 runs against radiosonde observations in the tropical 200–300 hPa layer over 1958–2017. The attribution literature does not identify any specific model as having reproduced the observed record; it invokes "*the models*" in the collective and reports their ensemble mean.

6.3 The multi-model ensemble mean is not a statistical estimator

The attribution literature treats the multi-model ensemble mean and its inter-model spread as if they were sample statistics from a well-defined probability distribution. They are not. The individual CMIP integrations are neither independent nor identically distributed. They are not draws from any specifiable population. The modeling groups' choices of parameterization, tuning target, structural formulation, and initialization are neither random nor exchangeable. The Central Limit Theorem does not apply, and no theorem of mathematical statistics licenses treating the ensemble mean as an unbiased or consistent estimator of any physical quantity. Averaging outputs of a heterogeneous set of models, each with its own parameter and structural choices and none of which individually matches the physical record, into a "*mean*", and reporting the inter-model spread as a "*confidence interval*", does not

produce a statistical estimator of any physical quantity. The ensemble mean is an arithmetic average of model outputs. Its narrower spread relative to the individual model outputs is a numerical consequence of averaging, not a physical or statistical narrowing of uncertainty about the world.

6.4 The tolerance under which "*match*" is claimed is not physical match

Peer-reviewed comparisons of CMIP model output to atmospheric temperature observations report specific discrepancies at the factor-of-two level across every model run in the two most recent CMIP generations.

McKitrick and Christy (2018) [21], examining 102 CMIP5 model runs against radiosonde observations in the tropical 200–300 hPa layer over 1958–2017, report that all model runs warmed faster than the observations, and that the discrepancy is statistically significant in a majority of individual cases. The authors conclude that the observed pattern is inconsistent with the negative-lapse-rate feedback response to greenhouse forcing that the models embed.

McKitrick and Christy (2020) [22], examining 38 CMIP6 models against satellite, radiosonde, and reanalysis observations in the global and tropical lower- and mid-troposphere over 1979–2014, report that every one of the 38 model runs overpredicts warming in every target observational analog, in most cases significantly, and that the average model-observation trend difference is statistically significant.

Zou et al. (2023) [44] present the NOAA STAR v5.0 satellite-derived mid-tropospheric temperature record for 1979–2021, reporting a global mean TMT trend of 0.092 ± 0.043 K/decade and a total tropospheric trend (with stratospheric-cooling contamination removed) of 0.142 ± 0.045 K/decade.[1] McKitrick and Christy (2020) [22] report the CMIP6 tropospheric-layer mean trends over 1979–2014: the low-ECS group of models has an overall mean trend of 0.21°C/decade, the high-ECS group 0.28°C/decade, and every one of the 38 model runs lies above every observational analog. The satellite-derived observed trend is a fraction of the CMIP6 group means.

An observation-model discrepancy in which the CMIP6 model-group mean tropospheric trends (0.21°C/decade for the low-ECS group, 0.28°C/decade for the high-ECS group; [22]) sit substantially above the peer-reviewed observed total-tropospheric trend (0.142 ± 0.045 K/decade; [44]), sustained across every one of the 38 CMIP6 model runs, is not a validated match. It is a systematic bias documented in the peer-reviewed literature. The literature's practice of describing such divergence as "*consistent with*" observation, or as "*within the ensemble range*" when the ensemble range is widened by including the extreme high-sensitivity model runs, substitutes a permissive linguistic convention for a physical criterion.

[1] The unit tags K/decade and °C/decade attached by the source authors to these area-weighted arithmetic means of intensive-variable tropospheric-layer readings inherit the status established in § 6.1: such means have no thermodynamic referent, and the unit tags carry no physical meaning. The numerical values in this section and in § 6.5 are reported as tagged by the source authors, with no endorsement of the tags.

6.5 The pattern of systematic divergence extends across the climate system

The tropospheric-warming discrepancies documented above are not isolated cases. Systematic and quantitatively significant model-observation divergences are documented in the peer-reviewed literature across multiple major subsystems of the climate. The following are representative cases.

Tropical Pacific sea surface temperature gradient: wrong sign of the forced response. Byrne et al. (2026) [4] assess all observed and modeled zonal SST gradient trends of 20 years or longer over 1870–2024 across five observational datasets and 14 CMIP6 large ensembles. The observed east-west gradient has strengthened at an increasing rate over the last 150 years. The CMIP6 forced response is a weakening of the gradient. The observed trend is not reproduced by any model's forced response; the few individual model runs that match the observations do so through internal variability that compensates for the model's gradient-weakening forced response. This is a sign disagreement between the forced response of the model ensemble and the observation.

Antarctic sea ice extent: wrong sign of the trend over 1979–2014. Shu et al. (2020) [38], examining 44 CMIP6 models against passive-microwave satellite observations, report that "*the observed weak but significant upward trend of the Antarctic SIE is not captured*" and that this "*was an issue already in the CMIP5 phase*." Roach and Polvani (2026) [31], in a paper titled "*Persistent Antarctic Sea Ice Biases in CMIP6 Models in spite of the Recent Decadelong Sharp Decline*" in the *Journal of Climate*, examine 462 ensemble members from 56 CMIP6 models and report that only 8 ensemble members (less than 2%) are consistent with the observed 1979–2014 expansion, and that even those simulations have climatological sea-ice biases (many under half of observed values) and variability and regional patterns that do not resemble observations. The authors conclude that "*not a single simulation among the several hundreds available from many models in CMIP6 resembles observations for both the 35-yr expansion phase and the 11-yr decline phase.*"

Arctic September sea ice: underestimated decline. Shu et al. (2020) [38] report that the observed Arctic September sea ice extent declining trend of -0.82 ± 0.18 million km^2 per decade over 1979–2014 is underestimated by the CMIP6 multi-model mean at -0.70 ± 0.06 million km^2 per decade, and that CMIP6 models do not reproduce the observed post-2000 acceleration of the decline.

Northern Hemisphere spring snow cover: CMIP5 underestimation of observed decline. Brutel-Vuilmet et al. (2013) [3] report that CMIP5 models underestimate the observed spring Northern Hemisphere snow cover extent trend by a factor of approximately three over 1979–2005: observed $(-3.4 \pm 1.1)\%$ per decade, simulated $(-1.0 \pm 0.3)\%$ per decade. Connolly et al. (2019) [10], comparing 196 CMIP5 runs against satellite observations of Northern Hemisphere snow cover over 1967–2018, report that "*the climate models were found to poorly explain the observed trends*": the models predict decreases in all four seasons, while observations show decreases only in spring and summer.

Regional precipitation over the Americas: models miss magnitude and sign. Qiu et al. (2024) [30] report a statistically significant observed drying trend in southwestern North America and a statistically significant observed wetting trend in the Amazon over 1979–2014. Coupled CMIP6 historical simulations reproduce neither trend. The authors trace the failure to the coupled models' inability to reproduce the observed Pacific trade-wind

intensification: the models simulate a weakened Walker circulation and an El Niño-like SST pattern, delivering the opposite regional precipitation response.

Each of these cases involves a systematic, statistically significant divergence between the multi-model ensemble and the peer-reviewed observation, documented by researchers who work within the climate-modeling community and endorse the general enterprise of climate simulation. Several of the divergences are sign disagreements, not merely magnitude disagreements. The pattern is a documented systematic divergence between model output and observation across multiple major climate subsystems.

6.6 Consequence for the defense

The factual-reproduction defense fails at every level. Its comparison target is not a physical quantity (§ 6.1). Its individual models, examined systematically in McKitrick and Christy (2018, 2020) [21, 22], all overpredict warming against every observational analog (§ 6.2). Its ensemble-mean object is not a statistical estimator (§ 6.3). The CMIP6 model-group mean tropospheric trends sit substantially above the peer-reviewed observed value ([22]; [44]), and analogous divergences of comparable or greater magnitude, including sign disagreements, are documented across the tropical Pacific SST gradient, Antarctic sea ice extent, Arctic September sea ice, Northern Hemisphere spring snow cover ([3]; [10]), and regional precipitation over the Americas (§ 6.4, § 6.5). Step 5 of the proof of Theorem A.11 shows that even if the defense's premise were granted, the consequent would not follow. This section shows that the premise is separately empirically false in its own terms across multiple major climate subsystems. The defense fails both on identification grounds and on empirical grounds.

7 Conclusion

The extreme-event attribution ratio $R(E)$ is a ratio of two model-internal probabilities. The counterfactual sample space Ω_C on which the denominator would need a physical counterpart fails every operational-access condition of Definition A.1, as established by Lemma A.13. No procedure applied to observations of the physical world can validate the denominator as a physical probability. Reproduction of the factual record does not select a unique counterfactual value, as directly measured by the factor-of-nineteen disagreement between two peer-reviewed attributions of the same 2021 Pacific Northwest heatwave. And, as Section 6 establishes, the factual-reproduction premise on which the defense of the practice rests is itself empirically false: the comparison target is not a physical quantity, every one of 38 CMIP6 model runs overpredicts warming against every observational analog [22], the CMIP6 model-group mean tropospheric trends sit substantially above the peer-reviewed observed trend ([22]; [44]), the multi-model ensemble mean is not a statistical estimator, and the "*match*" invoked in the literature is a linguistic convention rather than a physical criterion. The attribution ratio is therefore a statement about the model, not about the physical world. Its presentation as a physical fact is a category error. This defect is prior to and independent of every technical defect of the underlying models and would remain even in the limit of perfect physics and perfect fit.

The specific misrepresentations cataloged in Section 3 are not isolated errors. They are the observed practice of the attribution literature and its downstream communicators, from

the World Weather Attribution rapid-analysis papers of Philip et al. (2022) [28], Zachariah et al. (2022a) [42], and Clarke et al. (2026) [7], through the IPCC Sixth Assessment Working Group I Chapter 11 assessment language [16], through the Climate Central and Carbon Brief communication channels, into national and international climate litigation. Section 4 proves, conditional on Theorem A.11, on the interpretive convention stated in Corollary A.18, on the professional-competence standard of Definition A.9, and on the empirical reliance-context premise underlying condition (iii), that this pattern satisfies the three conditions of scientific fraud stated in Definition A.8.

Theorem A.11 rules out any physical content for $R(E)$. A proven mathematical statement is not subject to revision by further empirical work. The sample space over which $P(E \mid C)$ would need to be a physical probability fails operational access under Lemma A.13, and no refinement of models, no improvement of parameterizations, no increase in ensemble size, no advance in computing power, and no methodological innovation can change that. The model-internal reading of $R(E)$, stated in Corollary A.18, remains available and is not touched by the theorem; the theorem rules out the physical reading only.

In the documented cases cataloged in Section 3, the presenters offer $R(E)$ in the physical reading, unqualified, to the public, to courts, to regulators, and to policymakers, with professional-competence access to the fact that the physical reading is unavailable to them, and in contexts of substantial reliance. Under Theorem A.11, the interpretive convention of Corollary A.18, Definition A.9, and the empirical reliance-context premise, that pattern is scientific fraud in the sense of Definition A.8. Section 4 verifies each of the three conditions of the definition against the pattern under those four premises.

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Conflict of Interest

The authors declare no conflict of interest.

Appendix A

Definitions

We work in the standard measure-theoretic framework of probability. The definitions of scientific fraud (Definitions A.8 and A.9) apply to any scientific claim, not only to attribution claims.

Definition A.1 (Operational access; physical probability)**.** A sample space Ω satisfies *operational access* when at least one of the following three conditions holds:

(OA-1) The elements of Ω are physically realized outcomes, and $P(A)$ for $A \subseteq \Omega$ can be estimated by the empirical frequency of A among those realizations.

(OA-2) The elements of Ω are outcomes of a repeatable experimental protocol whose repetitions are independently accessible to observation, and $P(A)$ can be estimated by the long-run relative frequency of A across the repetitions of that protocol.

(OA-3) The elements of Ω are outcomes of members of a well-defined class of independently-observed analog systems (for example, distinct copies of an apparatus, or distinct members of a population sharing the relevant causal structure), and $P(A)$ can be estimated by empirical frequency across the class.

A probability $P(A)$ is *physical* when Ω satisfies operational access. A credence-based probability is physical only when the credence is defined by reference to a sample space that satisfies operational access; a credence over an operationally-inaccessible sample space is a psychological attitude of the credence-holder, not a physical property of the world, and is not a physical probability in the sense of this definition.

Remark A.2. Operational access is the standard implicit in every experimental science. A fair-coin probability satisfies (OA-2). Life-table mortality rates satisfy (OA-3). Weather statistics over a stationary reference climatology satisfy (OA-1). What operational access excludes is the assignment of physical probability to sample spaces from which no observation is or can be drawn.

Definition A.3 (Model probability)**.** A probability $P_M(A)$ is *model-internal* when A is a measurable subset of a sample space Ω_M whose elements are outputs of a computational model M. The measure P_M is induced by the model's parameter distributions, initial-condition ensembles, and stochastic components.

Definition A.4 (Factual run)**.** The factual run F is a class of model integrations of M under an estimated anthropogenic-forcing history ϕ_F intended to approximate the observed forcing trajectory of Earth.

Definition A.5 (Counterfactual run)**.** The counterfactual run C is a class of model integrations of M under a forcing history ϕ_C constructed by removing the anthropogenic component from ϕ_F. The resulting trajectory class is by construction disjoint from any physically realized Earth history.

Definition A.6 (Attribution ratio)**.** For an event class E, the attribution ratio is

$$R(E) := \frac{P_M(E \mid F)}{P_M(E \mid C)}.$$

Definition A.7 (Epistemic type of a claim)**.** A scientific claim S has an *epistemic type* determined by the class of evidence that could in principle support it and the class of procedures that could in principle validate it. Distinct epistemic types include, without limitation:

(a) *Physical measurement*, a numerical result of an observation of a physical quantity under a defined procedure applied to a physical sample.

(b) *Empirical inference from measurements*, a claim obtained by statistical or deductive operations on physical measurements, whose validity is tied to the measurement chain.

(c) *Model-internal output*, a numerical result computed by a defined algorithm applied to defined inputs, whose validity within the model is a matter of computation and whose validity as a physical statement requires an independent argument.

(d) *Assumption*, a proposition posited without derivation from measurement or from prior derivations.

Claims of distinct epistemic type are not interchangeable. A claim of type (c) or (d) is not a claim of type (a) or (b), and its presentation as such is a misrepresentation of epistemic type.

Definition A.8 (Scientific fraud)**.** A published statement S is a *scientific fraud* when all of the following hold:

(i) S is presented to its audience as a claim of one epistemic type T_1, while S is in fact a claim of a distinct and weaker epistemic type T_2 (in the sense of Definition A.7).

(ii) The presenter of S has access to the information sufficient to establish that S is of type T_2 rather than T_1, in the operationalized sense of Definition A.9 below.

(iii) S is issued in a context in which the audience is expected to, or invited to, act on S under the belief that it is of type T_1, and such action carries material consequences for the audience or for third parties.

Intent to deceive is not a necessary element of scientific fraud in this definition. The offense consists in the material misrepresentation of epistemic type together with the presenter's access to the information that would correct it and the reliance context in which the misrepresentation is issued.

Definition A.9 (Professional-competence access)**.** A presenter of a numerical claim $C = f(M, \theta, D)$, where M is a computational model, θ is a parameter vector, D is an input data set, and f is a defined procedure computing C from (M, θ, D), is deemed to have *professional-competence access* to the fact that C is a claim of epistemic type (c) in Definition A.7, if and only if the presenter is competent to compute C or to interpret its computation. Competence to compute or interpret C entails knowledge of the identities of M, θ, D, and f; such knowledge is sufficient to establish that C is a function of M, θ, and D, and hence that C is a model-internal output rather than a physical measurement independent of M.

Remark A.10. Definition A.9 operationalizes the "or in the exercise of professional competence should have" language of ordinary fraud analysis. It requires that the presenter know how C was computed. The step from "C is computed as $f(M, \theta, D)$" to "C is of epistemic type (c)" is a definitional consequence of Definition A.7, not a further empirical premise.

The Theorem

Theorem A.11 (Counterfactual Referent Problem)**.** *The attribution ratio $R(E)$ is not a physical probability. It is a ratio of two model-internal probabilities. Its interpretation as a statement about the physical world, of the form* event E was made R times more likely by anthropogenic forcing*, is a category error, and this defect cannot be repaired by any procedure that operates on observations of the actual world.*

Remark A.12 (Generalization to other counterfactual attribution claims)**.** Theorem A.11 is stated for the frequency ratio $R(E)$ because that is the canonical form the attribution literature reports. Its argument is the failure of the counterfactual sample space Ω_C to support a physical probability. Any quantity whose definition requires evaluation on Ω_C, or whose truth value is defined by comparison between quantities on Ω_F and quantities on Ω_C, inherits the same defect. This includes counterfactual dependence claims ("*event E would not have occurred without anthropogenic forcing*"), counterfactual magnitude claims ("*temperatures are X °C warmer due to human influence*"; "*the storm was Y % more intense than it would have been*"), and detection claims about the effect of anthropogenic forcing on a specified physical event or event class. Corollary A.18 states the corresponding legitimate model-internal reading for each such claim.

Operational-Access Lemma

Before proving Theorem A.11, we establish the operational status of the counterfactual sample space.

Lemma A.13 (Operational inaccessibility of counterfactual Earth)**.** *Let Ω_C denote the sample space of counterfactual Earth trajectories under forcing ϕ_C. Then Ω_C fails each of the three operational-access conditions (OA-1), (OA-2), and (OA-3) of Definition A.1.*

Proof. We check each condition.

Failure of (OA-1). There is exactly one physically realized Earth trajectory. That trajectory is generated by the actual forcing history, which includes the anthropogenic component. It is therefore an element of the factual sample space Ω_F, not of Ω_C. No element of Ω_C is a physically realized outcome. Condition (OA-1) requires physically realized elements; it is therefore not satisfied.

Failure of (OA-2). Condition (OA-2) requires a repeatable experimental protocol whose repetitions are independently accessible to observation and would generate members of Ω_C. No such protocol exists. The industrial revolution cannot be reversed. No laboratory system reproduces the coupled atmosphere-ocean-cryosphere-biosphere dynamics of Earth at the required grid resolution and multidecadal integration time. The proposal to identify a climate-model integration under ϕ_C with an execution of an experimental protocol is a category confusion: a model integration is a computation, not an observation of a physical experiment, and its output is a member of Ω_M (the model-internal sample space of Definition A.3), not of Ω_C (the sample space of physical Earth trajectories). Condition (OA-2) is therefore not satisfied.

Failure of (OA-3). Condition (OA-3) requires a well-defined class of independently-observed analog systems sharing the causal structure of the sampled outcomes. No such

class exists for the Earth climate system. Earth's climate is a singleton on the observational record. Other planetary bodies in the solar system (Mars, Venus, Titan) and observed exoplanets differ from Earth in ordinal state variables (surface pressure, atmospheric composition, rotation rate, obliquity), in the presence or absence of oceans and biosphere, and in stellar forcing history, in ways that preclude their treatment as members of a common analog class for the purpose of estimating $P(E \mid C)$. Paleoclimate reconstructions cover the actual Earth trajectory under varying natural forcings; they do not populate Ω_C, which is defined by removal of the anthropogenic component of the recent forcing history from a system that in the paleoclimate epoch did not have that component to remove. Condition (OA-3) is therefore not satisfied.

All three conditions fail. Hence Ω_C does not satisfy operational access. ■ □

Remark A.14. The failure of (OA-1) follows from Definition A.1 and the singleton character of the realized Earth trajectory. The failures of (OA-2) and (OA-3) rest additionally on empirical premises about the physical world: that no experimental apparatus reproduces Earth's coupled climate dynamics at the required scale, and that no independently-observed planetary system constitutes a member of an Earth-analog class for the purpose of estimating $P(E \mid C)$. These premises are widely accepted and are supported by the state of experimental capability and of comparative planetology; they are not derived from Definitions A.1–A.7. A party wishing to reject the lemma must reject one of these empirical premises and identify the analog system or protocol it invokes in place of the one currently unavailable.

Corollary A.15 (Non-existence of a physical probability on Ω_C)**.** *No probability measure defined on Ω_C is a physical probability in the sense of Definition A.1, including any credence-based measure, since credence-based physical probability requires operational access to the underlying sample space, and Ω_C lacks it by Lemma A.13.*

Remark A.16 (Robustness to interpretive tradition)**.** Definition A.1's operational-access conditions state the requirement, standard in the frequentist tradition, that a physical probability be tied to a well-defined reference class of repeatable trials or their independently observable analogs. This is the requirement of scientific measurement: a probability without an operational referent is not a measurement of anything in the physical world. A subjectivist alternative in which $P(E \mid C)$ is treated as a rational agent's credence without operational access to Ω_C is internally consistent as a formal object but is not a physical statement. On the subjectivist reading, $P(E \mid C)$ is a fact about the modeler's credences and not a fact about Earth's climate under a counterfactual forcing history. Theorem A.11's conclusion, that $R(E)$ is not a physical statement about the world, therefore holds under both traditions: on the operational-frequentist reading because Ω_C fails operational access; on the subjectivist reading because $P(E \mid C)$ is by construction a claim about credences rather than about physical events. Corollary A.18's legitimate model-internal reading covers both cases.

Proof of the Theorem

The proof proceeds in five steps.

Step 1. The denominator is not a physical probability

The counterfactual run C is defined as a set of model integrations under the forcing history ϕ_C. The probability $P_M(E \mid C)$ is a measure over the model-internal sample space Ω_M conditioned on ϕ_C. By Definition A.3, $P_M(E \mid C)$ is a model-internal probability.

The candidate physical interpretation of $P_M(E \mid C)$ would identify it with a probability defined on Ω_C, the sample space of counterfactual Earth trajectories. By Corollary A.15, no probability defined on Ω_C is a physical probability. Therefore no interpretation of $P_M(E \mid C)$ as a physical probability succeeds. $P_M(E \mid C)$ is a model-internal probability without a physical-probability counterpart.

Step 2. The numerator is also model-internal

The factual run F is a class of model integrations under ϕ_F. The trajectory class produced by F is not identical to the observed Earth trajectory. It approximates it under the model's approximations. The elements of the sample space of $P_M(E \mid F)$ are model outputs, not observations of the physical world.

The physically realized Earth trajectory constitutes a single sample of the underlying process. A probability defined over a single sample is not a frequency in the sense of (OA-1) applied to the physical trajectory alone. The probability $P_M(E \mid F)$ is therefore model-internal, and $P_M(E \mid F)$ is not equal to any observed frequency of E on Earth. (Where sufficient stationarity is assumed over the historical record, an empirical estimate of the frequency of the event class in the factual world may be available; this estimate is distinct from $P_M(E \mid F)$ and does not enter the attribution ratio.)

Step 3. The ratio inherits the status of its terms

$R(E)$ is a ratio of two model-internal probabilities. It is therefore itself a model-internal quantity. Any statement of the form $R(E)$ *measures the physical effect of anthropogenic forcing on the probability of* E asserts that a model-internal ratio equals a physical quantity. By Step 1 and Corollary A.15, no physical quantity of the required form exists on Ω_C. The identification is therefore impossible: there is no physical target for $R(E)$ to equal.

Step 4. No observation can validate the denominator

Let $\mathcal{O}$ denote any set of observations of the physical world. By definition, $\mathcal{O}$ consists of measurements taken on the physically realized Earth trajectory, which is a factual trajectory. No element of $\mathcal{O}$ is a measurement of any counterfactual trajectory, because by Lemma A.13 no counterfactual trajectory is operationally accessible.

Consider any procedure V that takes $\mathcal{O}$ as input and produces a validation of $P_M(E \mid C)$ as output. For V to validate $P_M(E \mid C)$ as a physical probability, V must establish that $P_M(E \mid C)$ equals a physical probability defined on Ω_C. By Corollary A.15, no such physical probability exists. V therefore cannot establish the required equality. This holds for every V and every $\mathcal{O}$.

Step 5. Even granting the factual-reproduction premise arguendo, no unique counterfactual output is selected

A common defense is that if the model M reproduces the observed factual record, then its counterfactual output should be trusted. This step defeats the defense on identification grounds, granting the antecedent solely arguendo. The antecedent is separately and independently false as an empirical matter; that failure is set out in Section 6. The present step assumes the antecedent for the sake of exposing the failure of its consequent.

Let $M(\theta)$ denote a model with parameter vector θ. Let Θ_F denote the set of parameter vectors under which $M(\theta)$ reproduces the factual record to within some fixed tolerance. Let $q(\theta) = P_{M(\theta)}(E \mid C)$ denote the counterfactual output.

Underdetermination of the counterfactual output by observation. The factual record is finite. The parameter space of a climate model has thousands of degrees of freedom (parameterizations of subgrid convection, cloud microphysics, aerosol interactions, ocean mixing, land-surface exchange, and boundary-layer turbulence, each with multiple tunable constants). Membership in Θ_F constrains a subset of these parameters. The counterfactual output q depends in part on parameters that the factual record does not constrain. Whether q is in fact constant on Θ_F for a given M is a further empirical question. What is established by the parameter-count argument is that constancy is not entailed by observation-fit: the parameter subset constrained by the factual record is a proper subset of the parameter subset governing the counterfactual response, and therefore constancy of q on Θ_F would be a coincidence of the model's construction rather than a consequence of the fit.

Field-level non-uniqueness of published counterfactual outputs. The theoretical underdetermination is instantiated at the level of the field's practice. Two peer-reviewed attribution studies of the same physical event (the 2021 Pacific Northwest heatwave) both reproduce the factual record within their respective tolerances and both compute counterfactual event-frequency estimates. Philip et al. (2022) [28], using a multi-model ensemble analyzed at the Royal Netherlands Meteorological Institute, report the event to be at least 150 times less frequent under ϕ_C. Leach et al. (2024) [18], using operational weather-prediction forecasts issued by the European Centre for Medium-Range Weather Forecasts, report the same event to be at least 8 times [2–50] less frequent under ϕ_C. The central estimates differ by a factor of approximately 19. The studies use different models, different ensemble constructions, and different event definitions; the disagreement is at the level of the field, not within a single Θ_F. That is sufficient for the identification argument here, since the defense under refutation is that the community's practice of tuning to the factual record delivers a well-identified counterfactual output. The community's practice does not deliver such an output. Different practitioners tuning to the same factual record produce counterfactual outputs whose intervals do not overlap.

Any choice of a specific counterfactual value from the observed range therefore reflects constraints beyond those imposed by observation. Those constraints are the modeler's priors on parameter selection, on model structure, on ensemble construction, and on event definition. The counterfactual output is a function of those priors. It is not a function of the observations alone.

Conclusion of the proof

By Steps 1 through 3, $R(E)$ is a ratio of model-internal probabilities and is itself a model-internal quantity, for which no physical counterpart exists on Ω_C. By Step 4, no observation of the physical world can validate the denominator as a physical probability. By Step 5, even granting the factual-reproduction premise arguendo, reproduction does not select a unique value of the denominator, either in theory or in the observed practice of the field. Therefore $R(E)$ is not a physical probability. Its presentation as a physical fact is a category error. ■

Corollaries

Corollary A.17 (Independence from technical failures)**.** *Theorem A.11 holds regardless of whether the underlying model M is empirically adequate on the factual record, whether its parameterizations resolve subgrid physics, whether ensembles agree, whether events are selected without bias, or whether return-period tails are correctly estimated. The category error established here is prior. It rejects the interpretation of $R(E)$ as a physical fact under all combinations of these auxiliary conditions, including the limit in which all of them are resolved.*

Corollary A.18 (Retreat to model-internal claims)**.** *An attribution claim whose truth is defined by comparison between quantities on the factual sample space Ω_F and quantities on the counterfactual sample space Ω_C is a legitimate statement about the model when explicitly qualified to model M under parameter vector θ. This applies whether the claim is stated as a frequency ratio $R(E)$, as a counterfactual dependence claim ("*event E would not have occurred without anthropogenic forcing*"; "*event E is a consequence of anthropogenic warming*"), as a counterfactual magnitude claim ("*temperatures are X°C warmer due to human influence*"; "*the storm was Y% more intense than it would have been*"), or as a detection claim about the effect of anthropogenic forcing on a specified physical event or event class. It is not a statement about the physical world. Any public communication of such a claim that omits an explicit model-scope qualifier is subject to the interpretive convention that an unqualified statement whose grammatical subject is a physical event, physical quantity, or physical event class, offered in a scientific-communication context, refers to the physical object named and not to a specific model's representation of it. Under that convention, such communication presents a model-internal quantity as a physical quantity.*

Remark A.19. The interpretive convention invoked in Corollary A.18 is an empirical premise about how unqualified counterfactual claims about physical events, quantities, and event classes are read, not a derivation from Definitions A.1–A.7. It rests on three independent grounds. First, the grammar of the statements: a bare sentence of the form "*event E was R times more likely*," "*event E would not have occurred without anthropogenic forcing*," "*temperatures are X °C warmer due to human influence*," or "*we can detect the effect of climate change on specific weather events*" takes as its subject a physical event, a physical quantity, or a physical event class, and states a property of it; the qualifier "*in model M*" is grammatically absent and its absence is not inferred. Second, the specification of the audience: IPCC Summaries for Policymakers, World Weather Attribution reports, and national broadcast interviews are drafted for readers explicitly identified as non-specialists

whose task is policy formation or civic understanding, not model interpretation; a communication drafted for such an audience without a model-scope qualifier is not read as bearing one. Third, the treatment of attribution claims as scientific evidence in legal and regulatory venues (Section 3, Misrepresentations 6 and 11): courts and regulators admit attribution ratios and counterfactual dependence claims as facts about the world subject to Federal Rule of Evidence 702 and equivalent standards, not as model outputs subject to a different admissibility inquiry. The convention is a matter of the practice of the field and its audiences and can in principle be defeated by evidence to the contrary; the burden of producing such evidence lies with the party invoking a model-internal reading of statements the field itself communicates without qualifier.

Corollary A.20 (Non-falsifiability of the attribution claim)**.** *No observation of the actual world can falsify or confirm the physical interpretation of $R(E)$, because that interpretation concerns a sample space that fails operational access (Lemma A.13). In the sense of Popper (1959) [29], the physical attribution claim is not a scientific statement.*

Appendix B

We now apply Definition A.8 to the material cataloged in Section 3.

Proposition B.1 (Attribution communications are scientific fraud under stated premises)**.** *Conditional on (a) Theorem A.11, (b) the interpretive convention stated in Corollary A.18 and its accompanying Remark, (c) the professional-competence standard of Definition A.9, and (d) the empirical premise that the class of documented statements is issued in reliance contexts with material consequences, each of the thirteen misrepresentation patterns documented in Section 3 satisfies all three conditions of Definition A.8 and therefore constitutes scientific fraud under that definition.*

Proof. We verify the three conditions of Definition A.8 in turn for the class of statements documented in Section 3, which class consists of published attribution ratios and derived confidence-language statements issued without a model-scope qualifier.

Condition (i): the statement is presented as a claim of one epistemic type while being a claim of a distinct, weaker type. The statements in question are grammatically claims about the frequency of physical events under specified counterfactual conditions. Section 3 supplies verbatim quotations: "*virtually impossible without human-caused climate change*", "*150 times less common without human-induced climate change*", "*30 times more likely because of human-caused climate change*", and analogous phrasings. Under the interpretive convention of Corollary A.18, these are presented as claims of epistemic type (a) or (b) in Definition A.7, namely as physical measurements of, or empirical inferences from measurements of, the effect of anthropogenic forcing on the probability of specific events. That the statements might in principle be re-read by an informed technical audience as compressed reports of model-internal computations does not alter the presented type: the presented type is the type to which the intended audience is directed under the convention, and the intended audience (the public, courts, regulators, and policymakers) does not import the model-scope qualifier. By Theorem A.11 and Corollary A.15, each such statement is in fact a claim of epistemic type (c), namely a model-internal output. Condition (i) is satisfied.

Condition (ii): the presenter has access to the information sufficient to establish the correct epistemic type. By Definition A.9, professional-competence access is the standard, and the standard is competence to compute or to interpret the claim, not authorship of the specific code that computed it. The presenters of the documented statements divide into two classes. First, the authors of specific attribution studies whose $R(E)$ output is reported as a physical fact (Philip et al., Zachariah et al., Leach et al., Climate Central, the IPCC Chapter 11 assessment team, and Clarke et al.). In every such case the presenter has authored, or has directly used, the model code, the parameter files, the ensemble configuration, and the counterfactual forcing specification, and therefore possesses the identities of M, θ, D, and f in the sense of Definition A.9. Second, senior climate scientists who invoke or restate attribution or counterfactual-dependence claims in op-eds, peer-reviewed statistical analyses of extreme anomalies, media interviews, social-media posts, national broadcast interviews, public lectures, or expert testimony in federal litigation (Mann, Hansen, Hayhoe, Hausfather, Trenberth, Schmidt). Each has held a senior research position at a major climate-science institution and has authored or co-authored peer-reviewed work using the modeling and statistical machinery from which counterfactual attribution claims are derived. Each is competent, in the sense of Definition A.9, to interpret an attribution ratio $R(E)$ or a counterfactual event-dependence claim as a function of a model M, its parameters θ, and its input data D, and to recognize such a claim as of epistemic type (c). Definition A.9 is thereby satisfied for every presenter in the documented pattern, without requiring personal authorship of the specific code underlying each cited claim.

Condition (iii): the statement is issued in a reliance context with material consequences. This condition is an empirical premise about the use of the statements. The documented statements are issued through press releases, peer-reviewed articles cited by press releases, IPCC assessment reports intended for policymakers by design, and aggregator platforms whose stated purpose is to inform public and legal decision-making. The reliance context runs through three distinct channels, each independently documented. First, litigation: Setzer and Higham (2024) [37] documents the citation of attribution ratios in climate litigation, including cases in which substantial damages, statutory obligations, and constitutional rights are at issue; Misrepresentation 11 ([17, 40]) is a direct instance in which a court adopted a counterfactual attribution claim as material evidence. Second, federal regulation: the US Environmental Protection Agency's *Endangerment and Cause or Contribute Findings for Greenhouse Gases under Section 202(a) of the Clean Air Act* [11], which provides the legal basis for federal greenhouse gas regulations, rests on assessment reports whose content includes attribution language, and the ongoing federal debate over the finding's rescission and defense continues to frame attribution studies as decision-relevant scientific evidence. Third, formal scientific assessment: the National Academies of Sciences, Engineering, and Medicine report on attribution of extreme weather events [24] was commissioned to assess the state of attribution science for policy purposes; the US National Climate Assessment and equivalent national and international assessment products cite attribution numbers in policy planning contexts. Insurance regulators cite the ratios in rate-approval decisions. The reliance context and its material consequences are established as an empirical matter across all three channels. Condition (iii) is satisfied.

All three conditions of Definition A.8 are satisfied for each pattern documented in Section 3, conditional on Theorem A.11, on the interpretive convention of Corollary A.18, on

Definition A.9, and on the empirical reliance-context premise underlying condition (iii). Therefore each pattern constitutes scientific fraud under Definition A.8, on the joint acceptance of those four premises. □

Remark B.2. The proposition is stated conditional on four premises: Theorem A.11, the interpretive convention of Corollary A.18, Definition A.9, and the empirical reliance-context premise underlying condition (iii). The conditionality is proper. Rejection of any one of the four blocks the proposition while leaving the others intact. A party who rejects Theorem A.11 rejects the classification of $R(E)$ as a claim of epistemic type (c), and therefore rejects condition (i). A party who rejects the interpretive convention blocks the presentation-as-type-(a)-or-(b) step in condition (i) even while accepting the theorem. A party who rejects Definition A.9, or the identification of professional-competence access with the ability to compute $R(E)$, blocks condition (ii). A party who rejects the empirical reliance-context premise blocks condition (iii). The proposition does not close over any of these disputes; it holds only on the joint acceptance of all four premises. A counterparty who accepts all four is committed to the proposition.

References

[1] Angélil, O., Stone, D., Perkins-Kirkpatrick, S., Alexander, L. V., Wehner, M., Shiogama, H., Wolski, P., Ciavarella, A., & Christidis, N. (2018). On the nonlinearity of spatial scales in extreme weather attribution statements. *Climate Dynamics*, *50*(7–8), 2739–2752.

[2] Kabukuru, W. (2022, November 16). Nigeria floods 80 times more likely with climate change. *AP News*.

[3] Brutel-Vuilmet, C., Ménégoz, M., & Krinner, G. (2013). An analysis of present and future seasonal Northern Hemisphere land snow cover simulated by CMIP5 coupled climate models. *The Cryosphere*, *7*, 67–80.

[4] Byrne, H., Seager, R., & Smerdon, J. E. (2026). CMIP6 models cannot capture long-term forced changes in the tropical Pacific sea surface temperature gradient. *Nature Communications*, *17*, 142.

[5] McSweeney, R., & Tandon, A. (2024, November 18). Mapped: How climate change affects extreme weather around the world [Interactive attribution tracker]. *Carbon Brief*. Last updated 19 March 2026.

[6] Dunne, D. (2022, May 23). Climate change made India and Pakistan's 2022 early heatwave '30 times more likely'. *Carbon Brief*.

[7] Clarke, B., Zachariah, M., Davies, B., Hock, R., Immerzeel, W., Jha, R., Robinson, T. R., Keeping, T., Barnes, C., Otto, F., Pandey, S., Vahlberg, M., Uprety, M., Singh, R., Arrighi, J., Lord, A., Ganguly, S., Dhakal, M., Rodriguez, L., … Raju, E. (2026, September 15). *Rapid warming in the Himalaya exacerbates geohazard cascades beyond adaptation limits* (WWA scientific report No. 92). World Weather Attribution.

[8] Climate Central. (2024, October 7). *Hurricane Milton rapid attribution: Climate change and Gulf of Mexico sea surface temperatures* [Climate Shift Index Alert].

[9] Cohler, J. (2026). *The Physical Tether Theorem: A rigorous formal synthesis of representational measurement theory, operationalism, and classical thermodynamics* [Manuscript submitted for publication].

[10] Connolly, R., Connolly, M., Soon, W., Legates, D. R., Cionco, R. G., & Velasco Herrera, V. M. (2019). Northern Hemisphere snow-cover trends (1967–2018): A comparison between climate models and observations. *Geosciences*, *9*(3), 135.

[11] U.S. Environmental Protection Agency. (2009, December 15). Endangerment and cause or contribute findings for greenhouse gases under Section 202(a) of the Clean Air Act; Final rule. *Federal Register*, *74*, 66496–66546. Docket ID EPA-HQ-OAR-2009-0171.

[12] Hansen, J., Sato, M., & Ruedy, R. (2012). Perception of climate change. *Proceedings of the National Academy of Sciences*, *109*(37), E2415–E2423.

[13] Hausfather, Z. [`@hausfath`]. (2021, June 25). *This is a good way to think about climate change and extreme heat events. What was once a* [Post]. X. https://x.com/hausfath/status/1408509067585196036

[14] Hayhoe, K. (2020, August 30). How climate change is fueling hurricanes and wildfires [Interview with M. Martin]. *All Things Considered*. NPR.

[15] Herring, S. C., Christidis, N., Hoell, A., Hoerling, M. P., & Stott, P. A. (Eds.). (2022). Explaining extreme events of 2020 from a climate perspective. *Bulletin of the American Meteorological Society*, *103*(3), S1–S117.

[16] Intergovernmental Panel on Climate Change. (2021). *Climate change 2021: The physical science basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.

[17] *Juliana v. United States*, 339 F. Supp. 3d 1062 (D. Or. 2018).

[18] Leach, N. J., Roberts, C. D., Aengenheyster, M., Heathcote, D., Mitchell, D. M., Thompson, V., Palmer, T., Weisheimer, A., & Allen, M. R. (2024). Heatwave attribution based on reliable operational weather forecasts. *Nature Communications*, *15*, 4530.

[19] Mann, M. E. (2017, August 28). It's a fact: Climate change made Hurricane Harvey more deadly. *The Guardian*.

[20] Mastrandrea, M. D., Field, C. B., Stocker, T. F., Edenhofer, O., Ebi, K. L., Frame, D. J., Held, H., Kriegler, E., Mach, K. J., Matschoss, P. R., Plattner, G.-K., Yohe, G. W., & Zwiers, F. W. (2010). *Guidance note for lead authors of the IPCC Fifth Assessment Report on consistent treatment of uncertainties*. Intergovernmental Panel on Climate Change.

[21] McKitrick, R., & Christy, J. (2018). A test of the tropical 200- to 300-hPa warming rate in climate models. *Earth and Space Science*, *5*, 529–536.

[22] McKitrick, R., & Christy, J. (2020). Pervasive warming bias in CMIP6 tropospheric layers. *Earth and Space Science*, *7*, e2020EA001281.

[23] UK Met Office. (2022, May 18). *Climate change making heatwaves more intense* [Press release, author: G. Madge].

[24] National Academies of Sciences, Engineering, and Medicine. (2016). *Attribution of extreme weather events in the context of climate change*. National Academies Press.

[25] Oreskes, N., Shrader-Frechette, K., & Belitz, K. (1994). Verification, validation, and confirmation of numerical models in the earth sciences. *Science*, *263*(5147), 641–646.

[26] Otto, F. E. L. (2016). Extreme events: The art of attribution. *Nature Climate Change*, *6*, 342–343.

[27] Parker, W. S. (2020). Model evaluation: An adequacy-for-purpose view. *Philosophy of Science*, *87*(3), 457–477.

[28] Philip, S. Y., Kew, S. F., van Oldenborgh, G. J., Anslow, F. S., Seneviratne, S. I., Vautard, R., Coumou, D., Ebi, K. L., Arrighi, J., Singh, R., van Aalst, M., Pereira Marghidan, C., Wehner, M., Yang, W., Li, S., Schumacher, D. L., Hauser, M., Bonnet, R., Luu, L. N., ... Otto, F. E. L. (2022). Rapid attribution analysis of the extraordinary heat wave on the Pacific coast of the US and Canada in June 2021. *Earth System Dynamics*, *13*, 1689–1713.

[29] Popper, K. R. (1959). *The logic of scientific discovery*. Hutchinson.

[30] Qiu, W., Collins, M., Scaife, A. A., & Santoso, A. (2024). Tropical Pacific trends explain the discrepancy between observed and modelled rainfall change over the Americas. *npj Climate and Atmospheric Science*, *7*(1), 1–7.

[31] Roach, L. A., & Polvani, L. M. (2026). Persistent Antarctic sea ice biases in CMIP6 models in spite of the recent decadelong sharp decline. *Journal of Climate*, *39*, 1669–1679.

[32] Schmidt, G. A. (2021a, June 18). Let's talk about climate change [Interview]. *Gravity Assist* (Season 5, Episode 8). NASA.

[33] Schmidt, G. A. (2021b, September 29). NASA climate advisor: "We're seeing things happen ... that are clearly unprecedented" [Interview with G. Zee]. ABC News.

[34] Schmidt, G. A. [`@ClimateOfGavin`]. (2021c, June 25). *The potentially record-shattering extreme heat weather forecasts for the Pacific NW are pretty shocking, but some folks keep getting* [Thread]. X. Original account deleted; archived at https://threadreaderapp.com/thread/1408416536109404165.html

[35] Schmidt, G. A. (2021d, November 4). The urgent case for net zero. *The Spectator*.

[36] Schmidt, G. A. (2023, September 6). Climate scientist on extreme weather and long-term trends [Interview]. *PBS NewsHour*.

[37] Setzer, J., & Higham, C. (2024). *Global trends in climate change litigation: 2024 snapshot*. Grantham Research Institute on Climate Change and the Environment, London School of Economics.

[38] Shu, Q., Wang, Q., Song, Z., Qiao, F., Zhao, J., Chu, M., & Li, X. (2020). Assessment of sea ice extent in CMIP6 with comparison to observations and CMIP5. *Geophysical Research Letters*, *47*, e2020GL087965.

[39] Sirota, D. (2018, August 24). Michael Mann says this summer is a big warning about climate change. *Capital & Main*.

[40] Trenberth, K. E. (2018). *Expert report of Kevin E. Trenberth, Sc.D.* Juliana v. United States, No. 6:15-cv-01517-TC (D. Or.), Document 267-1, filed June 28, 2018, at 23. `https://bit.ly/Trenberth2018`

[41] Winsberg, E. (2018). *Philosophy and climate science*. Cambridge University Press.

[42] Zachariah, M., Arulalan, T., AchutaRao, K., Saeed, F., Jha, R., Dhasmana, M. K., Mondal, A., Bonnet, R., Vautard, R., Philip, S., Kew, S., Vahlberg, M., Singh, R., Arrighi, J., Heinrich, D., Thalheimer, L., Marghidan, C. P., Kapoor, A., van Aalst, M., & Otto, F. E. L. (2022a, May 23). *Climate change made devastating early heat in India and Pakistan 30 times more likely*. World Weather Attribution.

[43] Zachariah, M., Kimutai, J., Vahlberg, M., Barnes, C., Pinto, I., Winder, R., Kandima, C., Kandungure, C., Petty, C., Otto, F. E. L., Philip, S., Kew, S., Alfieri, L., Cascio Ingurgio, N., Salvi, A., Domeisen, D., Roman, C., Bausback, K., Singh, R., … van Aalst, M. (2022b, November 16). *Climate change exacerbated heavy rainfall leading to large scale flooding in highly vulnerable communities in West Africa*. World Weather Attribution.

[44] Zou, C.-Z., Xu, H., Hao, X., & Liu, Q. (2023). Mid-tropospheric layer temperature record derived from satellite microwave sounder observations with backward merging approach. *Journal of Geophysical Research: Atmospheres*, *128*, e2022JD037472.